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APPLIED RESEARCH

AEFFNet: Attention Enhanced Feature Fusion Network for Small Object Detection in UAV Imagery

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ABSTRACT The rapid advancement of unmanned aerial vehicle (UAV) technology has markedly increased the use of drone-captured imagery across various applications, necessitating enhanced accuracy and real-time performance in UAV image detection. Addressing the specific challenges posed by small and densely distributed objects in such images, we introduce an attention enhanced feature fusion network (AEFFNet) designed specifically for small object detection in UAV imagery. Firstly, a hybrid attention module with associated multi-axis frequency and spatial attention was designed to enhance the feature extraction of small objects. Secondly, an adjacent layer feature fusion module is innovatively proposed in order to boost the detection capabilities for small and occluded objects. Finally, a series experiments are conducted on the VisDrone2023 dataset, which involves a large number of small objects photographed by drones. Our evaluations, conducted on the VisDrone2023 dataset, demonstrate substantial improvements over the YOLOv8m baseline model, with a 3.0% increase in mean Average Precision (mAP) and a 4.4% rise in AP50.

INDEX TERMS Object detection, small object detection, attention mechanism, multi-scale feature fusion.

I. INTRODUCTION

In recent years, with the maturation of information technology and unmanned aerial vehicle (UAV) control systems, UAVs have found extensive applications in various fields such as photography, agriculture, urban surveillance [1], security [2], and more [3], [4]. Due to its cost-effectiveness and ease of deployment, UAV remote sensing imagery has swiftly become a crucial data source in the realm of low-altitude remote sensing research. As a result, achieving both accuracy and real-time performance has become a primary objective in UAV image recognition. However, several challenges need to be addressed to fulfill this mission. Firstly, the scale of objects can vary significantly owing to the changing altitudes and angles during UAV image capture. Secondly, occlusions between objects often occur due to the dense object presence in drone-captured images. Thirdly, drone-captured images may include confusing geographic elements due to their wide coverage. Lastly, the distribution density of objects can exhibit extreme variations. With the aforementioned challenges to tackle, object detection in drone-captured images poses a formidable quest. Nevertheless, advancements in technology and algorithm development are continuously working towards overcoming these obstacles, paving the way for improved UAV image analysis and applications in diverse fields.

To tackle the challenges outlined above, deep neural network (DNN) based detectors [5], [6], [7] have been employed in the realm of UAV image recognition, showing promising progress with the advancement of algorithms. However, the majority of widely used object detection models have been developed and trained primarily on

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datasets featuring natural scenes, such as MS COCO [8] and PASCALVOC [9]. Consequently, these models are inclined towards focusing on larger or medium-sized targets, which limits their performance when applied to images captured from the aerial perspective of UAVs. Recognizing this limitation, researchers have introduced various methods and models to enhance detector performance in UAV imagery. One notable approach is the Feature Pyramid Networks (FPN) [10], proposed in 2017, which merges low-level and high-level feature maps to bolster the network's capability in detecting small objects. This technique has since become a staple in multi-scale networks due to its effectiveness. A feedback-driven loss function was introduced to address the challenge of inadequate supervision on small objects, aiming to balance the loss distribution effectively in a study [11]. Furthermore, to improve the detection of small objects, several anchor-free methods like CornerNet [12], CenterNet [13], and FCOS [14] have been proposed. These methods are particularly beneficial since anchor-based models often prioritize larger targets, sometimes neglecting smaller objects. A significant advancement came with the introduction of TPH-YOLOv5 [15], which integrated a transformer prediction head into the YOLOv5 architecture. This addition enhanced detection accuracy by enabling the model to better understand contextual cues and object relationships within the images. Another noteworthy innovation is found in ASF-YOLO [16], where an attentional scale sequence fusion method was introduced. This method combines both spatial and scale information, further refining the model's ability to detect objects accurately across varying sizes and contexts.

Building upon the insights from previous studies, we judiciously opted for YOLOv8 as our foundational model. The YOLO series [17], [18], [19], [20], [21] holds a pivotal position among one-stage detectors, particularly chosen as the backbone for models tailored to small object detection. Our approach, rooted in the YOLO series framework, emphasizes multi-scale feature learning and contextual understanding, tailored specifically for real-time small object detection in UAV imagery. By harnessing the strengths of YOLO series algorithms and aligning them with the characteristics unique to UAV images, our algorithm aims to address limitations in UAV image recognition such as instances of missing and false detections. This is achieved through the enhancement of the detector's feature extraction capabilities and precise localization of small objects.

The innovation and contributions of our work are detailed as follows:

- A multi-axis frequency and spatial attention (MAFSA) module is proposed, enriching feature representation of boundary and location information through the integrated use of frequency and spatial attention. This module enhances the model's focus on target areas, reduces interference from complex backgrounds, and further improves the ability to detect small objects.
- The adjacent layer feature fusion (ALFF) module is designed. By adaptively learning the spatial weights of feature map fused at adjacent scales rather than simply element-wise sums or concatenation, the inconsistency between different feature scales is reduced and the detection accuracy is improved.
- An additional detection head is incorporated into the network's architecture, and all four detection head are reshaped with ALFF module. With the introduction of MAFSA module and other enhancements in the neck part, we build our attention enhanced feature fusion network (AEFFNet).

II. RELATED WORKS

A. OBJECT DETECTION

Based on their methodologies, current object detectors can be broadly classified into two groups: one-stage detectors, such as the YOLO series, SSD [22], DETR [23], and FCOS, and two-stage detectors, including Mask R-CNN [24], Cascade R-CNN [25], Faster R-CNN, CenterNet, among others. Generally speaking, one-stage detectors excel in speed and resource efficiency. These algorithms carry out both classification and regression in a single step, bypassing the need for candidate frame generation and treating location information as potential targets. On the contrary, two-stage detectors typically exhibit superior accuracy and localization capabilities. They begin by utilizing CNN networks to generate candidate regions and subsequently perform classification and positional regression on these selected regions. The choice between one-stage and two-stage detectors often hinges on the specific requirements of the task at hand. For scenarios where real-time processing and efficiency are paramount, one-stage detectors offer a compelling solution. Conversely, when utmost precision and fine-grained localization are crucial, two-stage detectors stand out as the preferred choice. This diversity in detector designs reflects the evolving needs and challenges within the domain of object detection, allowing researchers and practitioners to tailor their approaches to match the demands of diverse applications.

From a component perspective, both one-stage and two-stage detectors typically comprise two main components. The first component is a CNN-based backbone responsible for image feature extraction. This backbone often includes well-known architectures such as VGG [26], ResNet [27], EfficientNet [28], CSPDarknet53 [29], Swin-Transformer [30], and others. These networks have demonstrated strong feature extraction capabilities, and researchers often fine-tune them to adapt the backbone specifically for their tasks. The second component is the detection head, utilized for predicting the class and bounding box for detected objects. This particular component is what distinguishes detectors into one-stage and two-stage categories. Detectors belonging to the latter category predict the bounding box first and subsequently determine the class of the object. In contrast, one-stage detectors perform both tasks simultaneously. Additionally, researchers frequently insert intermediate layers between the backbone and the detection

head, forming a component often referred to as the ''neck'' of object detectors developed in recent years. The neck section is designed to optimize the utilization of features extracted by the backbone. It reprocesses and strategically leverages the feature maps obtained at various stages of the backbone network. The interplay between these components, the backbone for feature extraction, the neck for feature refinement, and the detection head for final object classification and localization, forms the foundational structure of modern object detection frameworks. By carefully designing and fine-tuning each component, researchers aim to achieve optimal performance across a range of object detection tasks and datasets.

B. ATTENTION MECHANISM

The attention mechanism, rooted in the visual attention of individuals, mimics the selective focus that humans exhibit when observing objects. This mechanism operates by assigning higher weights to prominent features and lower weights, or even zero weight, to weaker ones, aligning with the objectives of the task at hand. By doing so, the attention mechanism enables models to concentrate on the regions of interest within an image rather than processing the entire image indiscriminately, thereby enhancing the effectiveness and selectivity of data processing.

The application of the attention mechanism in computer vision traces back to the 1990s, where it was first utilized [31]. Subsequently, in 2014, the integration of the attention mechanism into Recurrent Neural Networks (RNNs) yielded remarkable results [32]. However, it was in 2017 with the introduction of the Transformer model that self-attention emerged as the most popular and widely discussed form of attention mechanism [33]. Over the years, the attention mechanism has found extensive applications across various domains such as natural language processing, object detection, image recognition, semantic segmentation, and more.

In the realm of object detection, the Squeeze and Excitation (SE) module, proposed by Hu et al. [34], allows networks to focus on the relationships between channels, automatically learning the importance of different channel features. Building upon this concept, Woo et al. introduced the convolutional block attention module (CBAM) [35], which integrates a spatial attention mechanism based on SE and employs maxpooling to enable the network to concentrate on the spatial regions of objects. Inspired by CBAM, Wang et al. devised the hybrid dilated convolution attention (HDCA) [36] model, emphasizing critical positions within images.

Through the integration of attention mechanisms, neural networks can autonomously learn and selectively focus on pertinent information within inputs, thereby enhancing the performance and generalization capabilities of the models. This capability has led to significant advancements in object detection systems, enabling them to better understand and interpret complex visual data, ultimately improving their accuracy and efficiency.

C. MULTI-SCALE FEATURE FUSION

Compared to larger targets, small targets in UAV images often present a challenge due to the limited number of available pixels, making it difficult to extract detailed features. Additionally, as the network depth increases, the feature and location information of these small targets can become diluted or lost, posing a hurdle for accurate detection. This necessitates the incorporation of both semantic and representation information for effectively detecting small targets. The integration of these two types of information through multi-scale learning has proven to be an effective strategy for enhancing the detection performance of small targets.

Among the widely used multi-scale learning networks, the Feature Pyramid Network stands out. FPN introduces a bottom-up and top-down network structure to enhance features by combining features from adjacent layers. This approach allows the network to extract characteristics of various resolutions at different scales, providing appropriate feature representations for targets of different sizes. By fusing multi-scale feature information, the model can more accurately predict objects of varying sizes across different scales, leading to improved performance. FPN has successfully enhanced the detection capabilities of multi-scale targets with minimal computational cost and without significantly increasing parameters, but there still remains some challenges to address. These include issues such as a reduction in the ability to represent multi-scale features directly due to semantic gaps between different layers, the loss of feature information from the highest pyramid level during downsampling, and the neglect of Region of Interest (ROI) features from other layers since the ROI features of each layer are predicted independently. Building on the FPN concept, several models have been proposed to further enhance performance. For instance, PANet [37] incorporates additional bottom-up path aggregation modules within the FPN top-down structure, facilitating more efficient transfer of shallow information to higher layers. This reduces the number of convolutional layers required for information flow from high-level to low-level feature graphs. Another notable approach, NAS-FPN [38], employs neural architecture search to create new connections. While effective, this method is often computationally intensive and complex, making it challenging to implement and interpret. BiFPN [39], on the other hand, improves upon the PANet connection scheme, enhancing efficiency and introducing learnable weights to determine the importance of different input features. This achieves a better balance between efficiency and computational cost.

In recent years, super-resolution techniques have been increasingly integrated into the domain of small object detection, often in conjunction with FPN, offering novel methodologies and insights. For instance, J. Noh et al. introduced an advanced high-resolution feature extractor that enhances the feature representation of small objects through hierarchical super-resolution techniques [40]. Similarly, Deng et al. proposed the Enhanced Feature Pyramid Network

(EFPN), which incorporates ultra-high-resolution pyramid layers specifically designed for small object detection [41]. Furthermore, Yang et al. developed QueryDet [42], a framework that employs an innovative query mechanism to accelerate the inference speed of feature pyramid-based object detectors, thereby leveraging the benefits of high-resolution feature maps while avoiding computationally expensive processing of background regions.

III. METHODS

A. NETWORK ARCHITECTURE

The proposed model comprises a feature extraction network, an attention-enhanced adaptive multi-scale feature fusion network, and four detection heads. The core concept of the algorithm is to emphasize the retention of detailed feature information crucial for UAV image object detection, while also prioritizing the real-time performance of the model to the greatest extent possible. Within the neck and detection head section, we have designed the multi-axis frequency and spatial attention module and the adjacent layer feature fusion network to specifically target the detection of small objects. These additions, including the MAFSA module and ALFFNet, significantly enhance the performance of the benchmark model for small object detection within UAV images. While there is an increase in computational cost, this design enables the model to effectively focus on object information even in complex scenarios.

The feature extraction network is based on CSPDarknet53, serving as the backbone of the network. To ensure robust feature extraction capabilities without unnecessarily increasing model complexity, the backbone network undergoes adjustments with SPD [43] and Vanilla Blocks [44] while maintaining its core structure. This structure includes several convolutional layers and residual structures based on CSPNet, integrating convolutions of various kernel sizes to fuse information through feature connections. These structures exchange image feature information across different channels without altering the size of the feature map. The fundamental convolution module, comprised of convolutions, batch normalization (BN), and sigmoid-weighted linear units (SiLUs), forms the Convolution blocks within our network. This module allows for the comprehensive utilization of the extracted information from the backbone network. To further leverage the extracted information, the architecture of the neck is enhanced, drawing inspiration from BiFPN. This includes the incorporation of extra connections and adaptive fusion factors, which effectively enhance the network's ability to express features. Our MAFSA module is also fused into this part, introducing attention on both frequency and spatial domains.

Given that small objects in UAV images may only occupy one or two pixels in the deep feature map, it is crucial to prevent the loss of feature information for these tiny targets. To address this, a detection head, denoted as P2, is added to the head section. P2 performs object prediction on a feature map that undergoes three times up-sampling, resulting in a size of 160 × 160. This enables the detection of extremely small-scale objects. Consequently, our detection head incorporates four feature layers of dimensions 20 × 20, 40 × 40, 80 × 80, and 160 × 160, respectively. Furthermore, in each detection head, we include a ALFF module to fuse on channels rich in information and spatial features relevant to small objects. This integration enhances our model's ability to accurately identify and locate small targets within images. The overall structure of our proposed model is depicted in Fig.1, and its simplified edition is shown in Fig.2.

B. NECK STRUCTURE ENHANCEMENT

The original model's neck section, based on PANet, incorporates both a top-down and a bottom-up path for feature fusion across different layers. To bolster our network's detection capabilities for small objects, we introduced an additional output path in the P2 layer, along with two additional connections in the P3 and P4 layers. The extra output path in the P2 layer aims to capture finer details. For instance, assuming an input image size of 640 × 640, the P2 feature map size is 160 × 160. This means each feature map corresponds to a 4 × 4 receptive field in the input map, enhancing the detection of small objects and providing more detailed information for other layers during feature fusion. However, as the P2 layer is only down-sampled by a factor of four, it contains more interference information compared to the P3 and P4 layers. To address this, we have improved the feature fusion structure for enhanced feature extraction.

Inspired by the BiFPN concept, we introduced an additional edge in both the P2 and P3 layers seperately. This bidirectional feature fusion structure incorporates different scales of contextual information and balances weight information based on PANet. This system enables efficient feature fusion across various layers without significant computational overhead, simplifying multi-scale feature fusion while achieving a larger receptive field, precise object localization, and rich semantic information. The comparison between the original neck part structure and our enhanced model is illustrated below (Fig.3). After adding the P2 layer, an up-sampling process is introduced to fuse with the P3 layer, along with the addition of an up-sampling node to the P3 layer. Moreover, including pathways to the output nodes in the P3 and P4 layers enables efficient feature fusion without introducing excessive complexity.

In the original method, the features extracted at different levels were treated uniformly, undergoing a simple addition and fusion process. However, in this article, we have adopted a different approach by assigning learnable weights to these features, thereby distinguishing and emphasizing the importance of each weight. This allows the model to dynamically adjust the contribution of features from different levels, leading to a more refined and adaptive feature fusion process. The weighting method is as Eqs. (1) and (2):

$$O = \sum_i \omega_i \cdot I_i \tag{1}$$

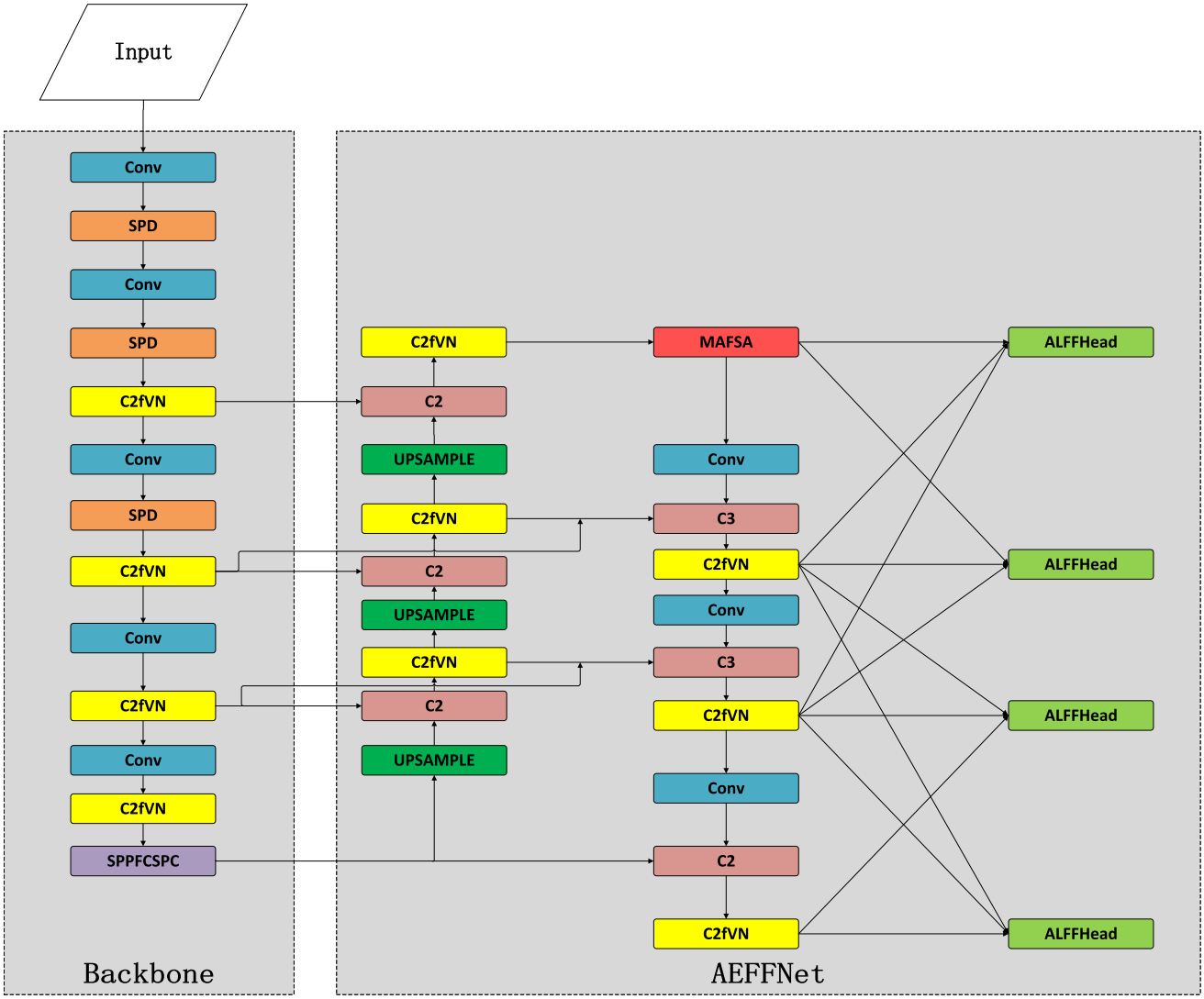


FIGURE 1. The complete architecture of our model. The backbone consists of CSPDarkNet53, along with the introduced SPD module and the Vanilla Block.The MAFSA module is introduced into the neck part, together with some extra edges and layers. In the head part, an additional detection head specially for small target is added, and all four heads are reshaped with the ALFF module.

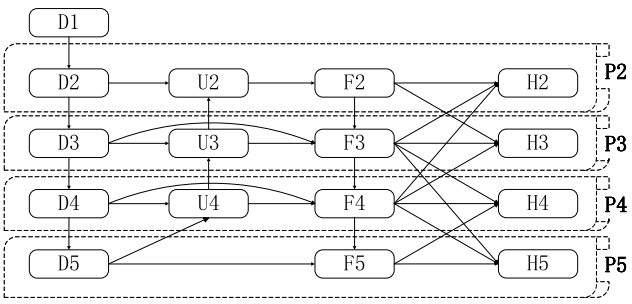


FIGURE 2. A simplified edition of complete architecture. In this figure, D represents the feature map obtained through downsampling, U represents that obtained through upsampling, F represents that used for detecting, H means detection head, and the numbers correspond to levels.

In this equation, ω_i represents a learnable weight, which can correspond to either a pixel or a channel of information. Since the scalar weights are not inherently bounded, normalization is necessary to ensure that each weight falls within the range of 0 and 1. To address this, we normalize the weights using the following method:

$$O = \frac{\sum_i \omega_i}{\sum_j \omega_j + \epsilon} \cdot I_i \tag{2}$$

where I_i represents the input value, while O is the output value.

C. ADJACENT LAYER FEATURE FUSION

In essence, the feature maps found in the deeper layers of the network are well-suited for classification tasks due to their large receptive field and rich semantic information. However, the downside is that these deeper layers often have lower resolution, leading to a loss of fine-grained details.

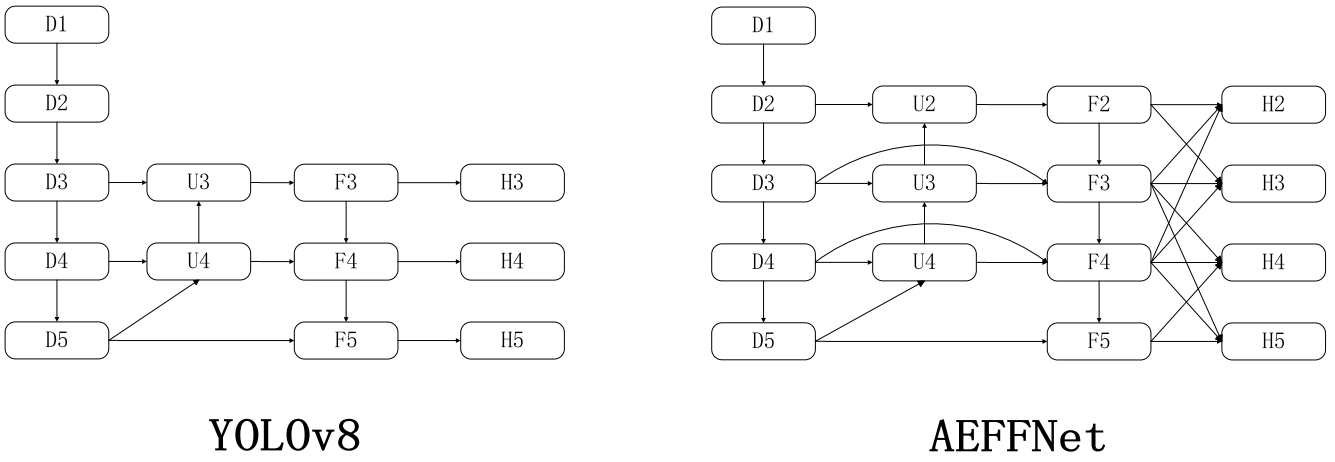


FIGURE 3. Comparison of feature fusion structures between YOLOv8 and AEFFNet.

Conversely, a shallow network with a smaller receptive field, higher resolution, and more detailed information is ideal for precise target localization. By fusing both shallow and deep features, the network is able to leverage the strengths of each level. This comprehensive approach enables the network to extract features from all levels, thereby enhancing feature extraction and acquiring a broader range of feature information, which includes robust semantic information from the deeper layers and finer details such as edges and textures from the shallower layers.

In contrast to previous approaches that typically integrate multi-level features through element-wise sum or concatenation, our novel approach involves adaptively learning the spatial weights for feature map fusion at adjacent scale. The pipeline for this method comprises two key steps: identically rescaling and adaptively fusing. Specifically, in the P3 and P4 layers, feature maps from one deeper layer and one shallower layer are fused into a local layer. This fusion process aims to incorporate both semantic and detailed information. Conversely, in the case of feature maps from two deeper layers or two shallower layers, they are fused into the P2 or P5 layer. This adjustment is made because the former typically lacks precise semantic information, while the latter benefits from the additional details for the classification task.

We denote the features at resolution level l (where $l \in \{1, 2, 3, 4\}$ for AEFFNet) as x^l. To achieve this, for each level l, we resize the features x_n from the other level n (where $n \neq l$) to match the shape of x_l. Since the features at different levels possess varying resolutions and channel numbers, we have adapted our up-sampling and down-sampling strategies accordingly for each scale. For up-sampling, we first utilize a 1×1 convolution layer to reduce the number of channels of the features to match that of level l, and subsequently we upscale the resolutions of these features using interpolation techniques. For down-sampling with a 1/2 ratio, we employ a 3×3 convolution layer with a stride of 2. This approach simultaneously adjusts both the number of channels and the resolution of the features. In the case of a scale ratio of 1/4, we introduce a 2-stride max pooling layer before applying the 2-stride convolution. This sequence of operations helps to modify both the channel numbers and the resolution effectively.

In the adaptive fusion part, suppose the target level is l and the feature maps are from other levels, which is represented by n_1, n_2 and n_3, then the features are fused at the corresponding level as follows:

$$y^l = \alpha^l \cdot x^{n_1 \to l} + \beta^l \cdot x^{n_2 \to l} + \gamma^l \cdot x^{n_3 \to l} \tag{3}$$

where y^l implies the output feature maps, α^l, β^l and γ^l refer to the spatial importance weights for the feature maps at three input levels to level l, which are adaptively learned by the network. The latter three parameters are defined using the soft-max function with separate control parameters. This ensures that their sum is 1 and they fall within the range of 0 and 1. We utilize several 1×1 convolution layers to compute these weight parameters individually, which approach allows the parameters to be learned through standard back-propagation during training. With this method, the features at all the levels are adaptively aggregated at each scale, and the final outputs are used for object detection. To provide a more comprehensive illustration of the ALFF mechanism, we have included a detailed schematic of the ALFFHead architecture in Fig.4, using H3 as a representative example. In this schematic, US denotes the upsampling operation, DS represents the downsampling operation, while α, β, and γ correspond to their respective learnable coefficients, which are optimized during the training process to achieve optimal feature fusion across different scales.

D. MAFSA

In drone-captured images, the delineation between segmented objects and the background can sometimes be unclear in the spatial domain due to occlusions between objects

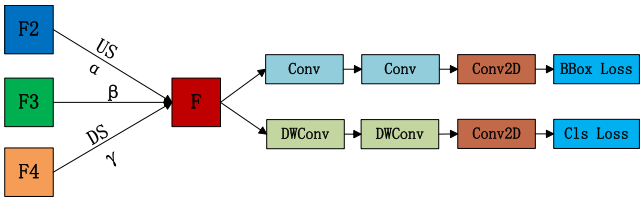


FIGURE 4. The architecture of ALFFHead.

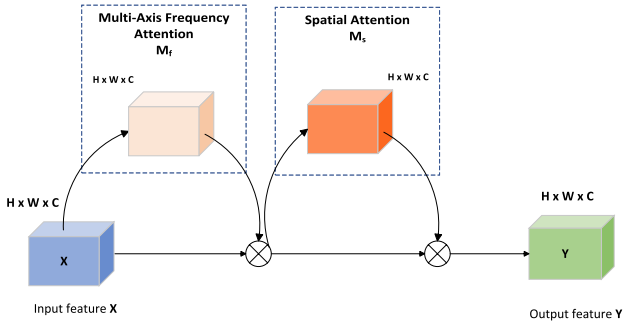


FIGURE 5. Multi-axis frequency and spatial attention module.

and confusing geographic elements in the images. However, in the frequency domain, objects typically exhibit distinct frequencies and focusing on specific frequency bands can enhance relevant feature detection. Therefore, incorporating frequency domain information using the attention mechanism aids our model in achieving more accurate and clearer segmentation. This approach helps reduce the influence of background and noise, thereby enhancing the segmentation performance. To achieve this goal, we have introduced a novel attention mechanism called the multi-axis frequency and spatial attention module into the P2 level, aimed at enhancing our model's performance in segmenting small targets. The MAFSA module comprises a sequence of a multi-axis frequency domain attention and a simplified spatial attention module, a novel architecture that leverages both spatial and frequency domain processing to enhance feature extraction while maintaining computational efficiency. The overall architecture of this module is depicted in Fig.5.

1) FREQUENCY FEATURE ENHANCEMENT

In Fig.5, M_f, M_s in the figure are weights of multi-axis frequency attention and spatial attention maps, respectively, and $\otimes$ denotes multiplication by element. The module first adopts multi-axis frequency attention to adjust the adaptive weights of the frequency dimension of the feature map, obtaining more comprehensive and rich information, thus, multi-directional structure and texture information, while effectively suppressing irrelevant noise and interference. In this way the feature fusion can focus on more details and valuable features. An input feature map X of $C \times H \times W$ with C, H and W representing the channel, height, and width of the map is considered. X is initially divided into four equal parts along the channel dimension, which are subsequently sent

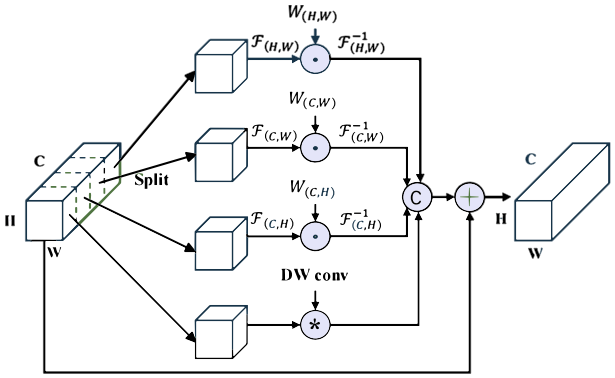


FIGURE 6. Multi-axis frequency attention module.

into four different branches x_1 - x_4, and these four parts are handled separately. This process is expressed by the following four equations.

$$x_1, x_2, x_3, x_4 = Split(X) \tag{4}$$

$$x_{i(I,J)} = W_{(I,J)} \odot \mathcal{F}_{(I,J)}[x_i] \tag{5}$$

$$x'_i = \mathcal{F}^{-1}_{(I,J)}[x_{(i(I,J))}], \quad x'_4 = DSC(x_4) \tag{6}$$

$$Y = Concat(x'_1, x'_2, x'_3, x'_4) + X \tag{7}$$

for $i = 1,2,3$ each number corresponds to one of the first three branches respectively. The terms $W_{(I,J)}$ and $\mathcal{F}_{(I,J)}$ represent the learnable external weight and the 2-dimensional discrete Fourier transformation (2D DFT) for the respective axes. Specifically, when $i = 1$, the axes I and J refer to Height-Width; when $i = 2$, they refer to Channel-Width; and when $i = 3$, to Channel-Height. The symbol $\odot$ denotes the element-wise product. The term $\mathcal{F}^{-1}_{(I,J)}$ refers to the 2-dimensional inverse discrete Fourier transformation (2D IDFT). The operations *Split* and *Concat* involve splitting and concatenating along the channel dimension, respectively. The *DSC* means depthwise separable convolution.

In the first branch, the feature map undergoes a transformation to the frequency domain via a 2D DFT performed on the Height-Width axes. This is followed by an element-wise multiplication with the corresponding learnable weight, which is generated through a series of inverted depthwise separable convolutions (IDSC). The feature map is then converted back to the spatial domain using a 2D IDFT. Similarly, the second and third branches execute these operations on the Channel-Width and Channel-Height axes, respectively, allowing the model to capture more comprehensive global information. Additionally, local information is critical for the task of segmentation. Hence, in the fourth branch, depthwise separable convolutions (DSC) are used to extract local features. The feature maps from all four branches are then concatenated along the channel dimension, restoring them to the same size as the input. Finally, a residual connection to the input is employed to produce the output, enhancing the overall feature integration. The structure of this part is illustrated in Fig.6.

TABLE 1. Distribution of labels across categories, including count and proportion.

Category	Motor	Bus	Awn-T*	tricycle	Truck	Van	Car	Pedestrian	People	Bicycle
Quantity	29,646	5,926	3,246	4,812	12,875	24,956	144,866	79,335	27,059	10,480
Proportion (%)	8.64	1.73	0.95	1.4	3.75	7.27	42.21	23.12	7.88	3.05

*:Awn-T represents awning-tricycle

2) SPACIAL FEATURE ENHANCEMENT

Spatial feature enhancement is used after frequency feature enhancement using the spatial attention for spatial features from the spatial dimension, embedding spatial position information into the frequency weights and filtering out channels that are sensitive to location information during fusion. This module captures the important spatial areas of feature maps via incorporating attention mechanism in the domain of space.

Specifically, assume that the output feature map, F_2, from the multi-axis frequency attention module, has dimensions of $C \times H \times W$. This map is processed by a sequence of layers that includes two convolution layers with a large kernel (kernel size of 7) that followed by a batch normalization layer separately and a rectified linear unit (ReLU). Subsequently, a sigmoid layer generates weights, which are then element-wise multiplied with the original feature map, F_s, to produce the final feature map, F_3.

IV. EXPERIMENT AND RESULTS

This section describes the dataset and experimental circumstances used to evaluate the performance of our algorithm against those of other state-of-the-art approaches. Finally, we conduct a series of ablation experiments and visualization of experimental results.

A. DATASETS

We evaluated our model using the VisDrone2023 dataset, which was curated by the Tianjin University Machine Learning and Data Mining Laboratory team. This comprehensive benchmark dataset comprises images captured by drones across 14 different cities in China, offering a diverse range of object densities and a complex environment for aerial drone photography. It stands as one of the most extensive datasets with broad coverage in China. The dataset includes a total of 10,209 still images: 6,471 for training, 3,190 for testing, and 548 for validation. The image dimensions in the dataset are approximately 2000 × 1500 pixels. The dataset covers ten categories of objects: cars, pedestrians, buses, bicycles, tricycles, motorcycles, awning-tricycles, trucks, vans, and people, totaling 2.6 million labels. The VisDrone2023 dataset accurately reflects real-world UAV appli0cations and aligns perfectly with the focus and objectives of this paper. Notably, this dataset has a significant proportion of small targets, making it ideal for our real-time testing experiments and ablation study. The specific quantities and proportions are detailed in Table 1.

B. ENVIRONMENT AND DETAILS

Our proposed model is implemented using the PyTorch_2.1.0 deep learning framework, with Python_3.10.13 as the programming language, running on a machine equipped with an NVIDIA GTX 4070Ti GPU. We maintain consistent hyperparameters throughout the training, validation, and testing phases. The optimizer selected is Lion, with a momentum of 0.9 and a weight decay set to 5×10^{-3}. To ensure stable training, we employ a warm-up strategy for the first three epochs among all the 300 epochs, and the image size is set to 800 × 800 with a batch size of 2. Additionally, the learning rate attenuation coefficient (gamma) for the aforementioned datasets is set to 0.01.

To assess the performance of the network, we utilize the following metrics: average precision (AP), parameters (Params), precision (P), giga floating point operations per second (GFLOPs), and frames per second (FPS). AP is a key evaluation metric for measuring the accuracy of object detection models. It represents the average value of precision for a range of recall levels. GFLOPs measure the computational complexity of training a model, quantifying one billion floating-point operations per second. Another important metric is the Params value, which indicates the total number of parameters in a model. This metric is crucial for assessing the consumption of computational memory resources. FPS measures how many images the model can process per second, serving as an indicator of the model's real-time performance. There is a direct relationship between FPS and the resolution of the images being processed: generally, the higher the resolution of the input image, the lower the FPS, assuming the model and operating environment remain constant.

C. RESULTS AND COMPARISON

In this section, we assess our model, AEFFNet, with the input image size set to 800 × 800. We compare its performance against the baseline YOLOv8m and other existing methods using the VisDrone2023 dataset. Table 2 displays the Mean Average Precision (mAP) for 10 categories, as well as the overall AP50 and mAP values. The results demonstrate that AEFFNet significantly outperforms the models submitted by the VisDrone team, especially in overall detection performance and across most individual categories. Specifically, AEFFNet shows improvements over TPH-YOLOv5, with increases of 5.6% in overall AP50 and 4.5% in mAP. The most substantial gain is observed in the motor category, where the mAP increased by 6.7%. Compared to the baseline YOLOv8m, AEFFNet achieved higher overall AP50 and

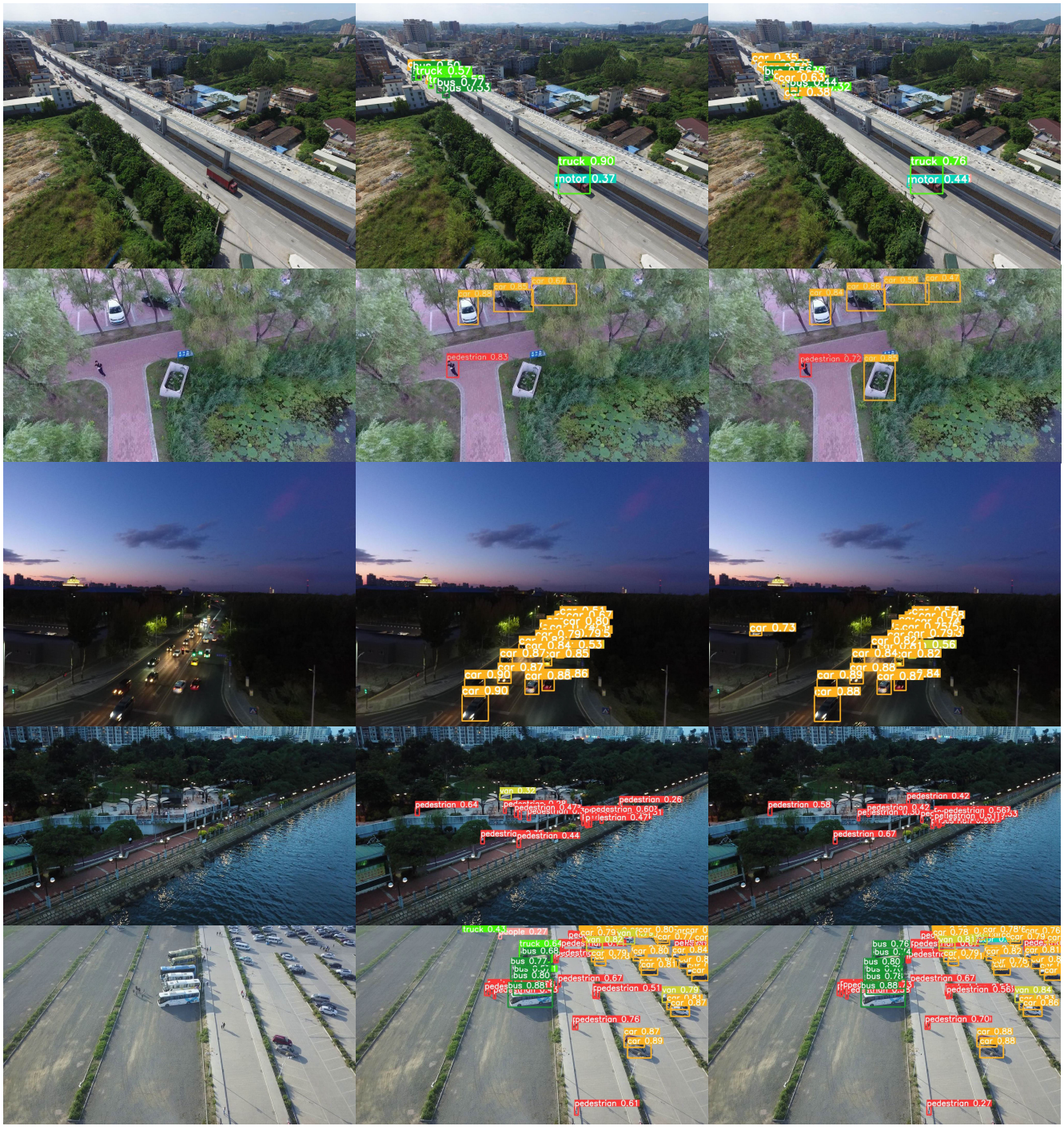


FIGURE 7. In the presented results from the VisDrone2023-DET-test-challenge, the first column shows the original image, the second column exhibits the detection outcomes generated by the YOLOv8m model, and the third column displays the results produced by the AEFFNet model. This arrangement allows for a direct visual comparison between the baseline model and the improved performance of AEFFNet.

mAP by 2.7% and 1.9%, respectively. The categories showing the most significant improvements include pedestrians, people, and awning-tricycle, with increases of 3.1%, 3.1%, and 2.5%, respectively. Overall, the performance analysis confirms that AEFFNet surpasses YOLOv8m in terms of detection accuracy. In comparison with YOLOv9, our model still shows advantages in overall performance and most categories except for truck and tricycle. YOLOv10 could also not match our model in accuracy since it rather more designed for faster and lighter-weighted.

While the AEFFNet model generally surpasses the baseline YOLOv8m, it underperforms in certain categories

TABLE 2. Comparison of methods on the VisDrone (%).

Method	AP50	mAP	PED	PEO	BIC	Car	Van	TRU	TRI	A-T	Bus	MO
FPN*	32.2	16.5	15.7	5.0	4.9	38.5	20.8	18.8	15.0	10.8	26.7	12.8
Light-RCNN* [45]	32.8	16.5	17.0	4.8	5.7	32.3	22.1	18.4	16.6	11.9	29.2	11.9
CornerNet*	34.1	17.4	20.4	6.6	4.6	40.9	20.2	20.5	14.0	9.3	24.4	12.1
CN-FaDhSa* [46]	49.5	28.5	30.5	12.9	9.9	52.5	38.1	32.9	25.9	**22.0**	39.9	20.6
RRNet* [47]	55.8	17.4	30.4	14.9	13.7	51.4	36.1	35.2	28.0	19.0	44.2	25.8
YOLOv6 3.0 [48]	47.6	29.0	25.4	16.1	11.0	59.0	36.5	32.4	22.1	12.4	49.3	25.2
TPH-YOLOv5l	48.1	29.2	27.3	16.9	10.8	61.8	37.4	31.9	20.8	12.4	47.6	25.1
TPH-YOLOv5++ [49]	47.2	29.0	28.7	17.8	11.5	61.3	36.2	31.0	20.3	12.0	45.4	26.2
YOLOv5 [50]	45.9	28.1	27.6	17.7	11.5	60.0	35.8	31.2	19.4	11.2	43.4	24.1
YOLOv8 [51]	50.1	31.4	28.5	20.0	13.2	63.4	38.4	33.1	23.5	13.5	51.4	28.9
YOLOv9 [20]	53.2	33.0	31.1	22.6	14.4	65.7	41.3	**35.8**	**26.9**	16.8	52.0	30.9
YOLOv10 [21]	49.6	30.8	27.6	19.4	12.0	64.1	40.8	31.9	22.1	13.0	52.1	29.4
AEFFNet	**54.5**	**34.4**	**33.1**	**23.2**	**15.2**	**66.1**	**42.4**	34.2	26.0	16.0	**55.5**	**32.5**

1. * indicates that the method is the algorithm result submitted by the VisDrone team, and the best results are marked in bold.
2. Some categories are expressed through abbreviations in the table: PED for pedestrian, PEO for people, BIC for bicycle, TRU for truck, TRI for tricycle, A-T for awning-tricycle, MO for motor.

TABLE 3. Ablation study of the proposed AEFFNet.

Methods	AP50 (%)	mAP (%)	Params (M)	GFLOPs	FPS
AEFFNet	54.5	34.4	32.91	142.3	68
AEFFNet - MAFSA	54.0	34.0	32.08	132.5	84.03
AEFFNet - ALFFN	52.9	33.0	23.77	118.3	76.48
AEFFNet - MAFSA - ALFFN	52.3	32.7	22.94	108.4	91.74
YOLOv8m	50.1	31.4	25.84	78.7	80.00

compared to methods submitted by the VisDrone team. Notably, the CN-FaDhSa algorithm achieves a 6.0% higher detection accuracy in the awning-tricycle category than our AEFFNet model. Detailed analysis suggests that this discrepancy in recognition accuracy may be attributed to the 800 × 800 input image size used during training. This resolution can lead to compression and loss of essential image details. Furthermore, the small size and limited number of targets within the awning-tricycle category present significant challenges for training, adversely affecting the model's detection performance in this specific area.

D. ABLATION STUDY

We assessed the impact of each component within the AEFFNet architecture using the VisDrone2023-DET-val dataset, as detailed in Table 3. To ascertain the contribution of each module to the overall performance, we systematically removed individual components from AEFFNet and evaluated their effects on the network's performance.

Additionally, to validate the efficacy of frequency-domain attention, we conducted a set of ablation experiments specifically for the MAFSA module. To eliminate interference from other factors, these experiments were based solely on the YOLOv8m model, to which we added spatial attention, frequency-domain attention, and the MAFSA module, respectively. The results demonstrate that this module enhances the model's accuracy in detecting small objects without significantly increasing the number of parameters or computational load. The details are in Table 4, where SA means spatial attention and FA means frequency attention.

TABLE 4. Ablation study of the MAFSA module.

Methods	AP50 (%)	mAP (%)	Params (M)	GFLOPs
YOLOv8m + MAFSA	51.0	32.0	26.35	84
YOLOv8m + FA	50.8	31.8	26.35	81.6
YOLOv8m + SA	50.4	31.5	26.35	81.1
YOLOv8m	50.1	31.4	25.84	78.7

E. VISUALIZE THE MODEL PREDICTIONS

At last, we present representative images from the VisDrone2023-DET-test-challenge dataset in Fig.7 to visually demonstrate the impact of AEFFNet. These images illustrate the model's enhanced detection capabilities, providing a clear comparison of its performance improvements over baseline models.

The results demonstrate that although AEFFNet occasionally missed or misclassified a few small objects, its detection capabilities for overlapping and extremely small objects have significantly improved compared to the baseline model. This highlights AEFFNet's enhanced precision and effectiveness in challenging detection scenarios.

V. CONCLUSION

Object detection in drone imagery presents considerable challenges. To improve the accuracy of such detections, we introduced the AEFFNet model, which incorporates the MAFSA and ALFFN modules to enhance performance. Experimental evaluations on the VisDrone2023-DET dataset showed that AEFFNet achieved a 3.0% increase in mAP and a 4.4% rise in AP50 compared to the baseline YOLOv8m model.

Despite these advancements, AEFFNet has limitations. These include high computational demands, suboptimal detection of extremely small or heavily occluded objects, and reduced FPS compared to the baseline. In our future work, we aim to refine AEFFNet's architecture to further improve recognition accuracy and reduce computational complexity, addressing these challenges more effectively.

DECLARATIONS

CONFLICT OF INTEREST

The authors declare that they have no conflict or interest in this article.

AVAILABILITY OF DATA AND MATERIALS

The VisDrone dataset that support the findings of this study are available from the website:
https://gitcode.com/visdrone/visdrone-dataset/overview.

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